Hemang Negandhi

Senior Data Engineer | BFSI Data Engineering | BI & Analytics | AI-Driven Data Systems

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PROFESSIONAL SUMMARY

BFSI-focused Senior Data Engineer with 8+ years in BFSI, building ETL/ELT pipelines, financial data processing, and reconciliation automation. Hands-on with Python and SQL for data engineering, and with Alteryx Designer, Power BI (DAX), and Tableau for analytics and reporting. Prototyping GenAI solutions with LLMs, RAG, and LangChain. MS in Computer Science, University of Texas at Arlington. Focused on reliable, well-governed data workflows that support decision-making in banking and financial services.

WORK HIGHLIGHTS

Project	What I Built	Impact
Financial Data Processing (BNP Paribas, C3 Project)	Process BFSI financial data monthly using Alteryx Designer, maintaining Data Governance and Data Quality standards.	**422 GB** processed monthly; BI insight for **10+** teams
Automated Financial Reconciliation (BNP Paribas, AXIOM–TRITON)	End-to-end reconciliation pipeline between AXIOM and TRITON using PowerShell, Alteryx, and Python; real-time regulatory reporting to Federal regulators via Power BI.	**900+ hours saved** annually
REST API Integration & Automation (AKA Enterprise Solution)	Dynamic analytics tool using the BambooHR REST API and Power BI; Data Quality improvements and HR Process Automation.	Adopted by **15 plus** stakeholders
ML Recommendation System (Master's Project, UT Arlington)	Scalable ML solution using Python, Flask, and GCP for predictive recommendations, applying NLP (stemming, lemmatization) and Feature Engineering.	**1.5M+** movie rows processed

RECENT AI WORK

Autonomous AI Job-Search Agent | Python, Groq API (LLaMA 3.3-70B), JobSpy, BeautifulSoup, Pydantic, Pandas, httpx

- Built an autonomous AI job-search agent in Python using JobSpy, REST APIs, and BeautifulSoup to scrape 50+ listings per run from LinkedIn, Indeed, and RemoteOK.
- Designed a multi-stage data pipeline with rule-based hard filtering, Pydantic validation, and URL-based deduplication — eliminating irrelevant results before LLM processing and reducing API token consumption by **~30%.**
- Engineered an LLM scoring engine using the Groq API (LLaMA 3.3-70B) with structured JSON prompt engineering and explainable 1–10 relevance scoring against a dynamic candidate profile.
- Automated end-to-end data output using Pandas to generate dated, score-sorted CSV reports.
- Applied AI agent design patterns — tool-using agent architecture, ReAct-style reasoning, cost-optimized LLM orchestration, and modular separation of scraping, scoring, filtering, and output layers — using Python, dotenv, and httpx.

TECHNICAL SKILLS

Languages: Python, SQL, PowerShell

Databases: PostgreSQL, MS SQL (T-SQL), Oracle 11g (PL/SQL), MySQL

Data Engineering: Alteryx, ETL/ELT, Data Warehousing, Data Modeling (Star/Snowflake Schema), SSRS, Analytics

Business Intelligence: Power BI (DAX), Tableau, Qlik Sense, QlikView, SSIS

AI / ML: Machine Learning, Generative AI, LLMs (Groq/LLaMA 3.3-70B), Prompt Engineering, RAG, LangChain, Agentic AI

Frameworks / Libraries: Flask, Apache Spark (PySpark), Pandas, NumPy, NLP (NLTK), Pydantic

Cloud Platforms: AWS (Redshift, EC2, S3), Google Cloud Platform (GCP), Microsoft Azure

Web: HTML, PHP, JavaScript

Methodology: CI/CD, Data Governance

Tools / IDEs: PyCharm, Visual Studio Code

WORK EXPERIENCE

Senior Data Engineer | BNP Paribas, Mumbai, India **Jul 2020 – Present**

- Analyze and design Alteryx workflows for ETL; perform performance tuning and query optimization in Tableau.
- Create Tableau Data Extracts (TDEs) for multiple projects and schedule them on Tableau Server per business requirements.
- Coordinate with Paris and Madrid teams to deliver business requirements and ad hoc requests.
- Maintain and enhance C3FINANCE, a core data source processing **422 GB** of data per month and delivering analytics to **10+** teams; improved data availability and reduced downstream query time by **30%**.

- Lead migration of the financial data hierarchy from FIREF to GRECO_PMA, increasing data granularity from **12 levels to 30 levels** and expanding insights by 25%.
- Spearhead the Deposit and Trend Analysis project, saving **786** work hours per year and reducing data errors from 10% to 2.5%; develop custom measures for financial forecasting and risk analysis, improving prediction accuracy by **25%**.
- Engineer an automated reconciliation pipeline between AXIOM and TRITON reporting systems using PowerShell (scheduled file detection every 10 minutes), Alteryx macros, and embedded Python; match records using a composite key of Date + Series + Version.
- Save **900+ hours** of manual work annually by automating the end-to-end reconciliation workflow with real-time email alerts at each stage and built-in error logging.
- Build a real-time regulatory reporting feed into Power BI from automated reconciliation outputs, enabling users to validate figures and submit accurate, timely reports to Federal regulators in the NAR region.
- Design version-aware, counter-based file detection logic to independently identify new or updated files on both drives, ensuring reconciliation triggers only on validated matched sets — improving accuracy by 75% and reducing errors from partial or duplicate file arrivals.

Business Data Analyst | BNY Mellon Pershing, New Jersey, USA **Sep 2019 – Mar 2020**

- Developed Alteryx workflows for data extraction, transformation, and loading (ETL).
- Used SQL to extract data and create ad hoc reports for visualization on Tableau Server.
- Conducted rigorous Tableau data validation by matching and verifying figures against the database, resulting in a 95% accuracy rate.
- Contributed to the development of Tableau workbooks for the EPO and Corporate Finance departments.
- Applied knowledge of star and snowflake dimensional schemas.
- Collaborated with cross-functional teams to gather business requirements and translate them into effective BI solutions, improving decision-making efficiency.

Power BI Intern | AKA Enterprise Solution, New York, USA **May 2018 – Jul 2018**

- Migrated an in-house application from a Power BI content workspace to an app workspace.
- Participated in creating dynamic visualizations to track employee performance.
- Established REST API integration with BambooHR to provide continuous access to real-time employee data; the tool was adopted by 15+ HR team members, improving efficiency and productivity.

Software Engineer | Infogain India Pvt Ltd, Mumbai, India **Aug 2015 – Jul 2017**

- Collected data in QlikSense from Oracle, CSV files, MS SQL, and MySQL sources.
- Created dynamic, interactive dashboards in Tableau, Power BI, and QlikSense.
- Analyzed underlying data for discrepancies, investigated errors, and implemented incremental loading.
- Performed ETL using SQL Server Integration Services (SSIS) and QlikView.
- Created a real-time streaming dashboard with real-time KPIs in Tableau.
- Documented ETL/ELT projects and workflows, including supporting documentation for troubleshooting.

Intern | CloudFronts Technologies, Mumbai, India **Jun 2015**

- Gained hands-on experience with SSIS, SSRS, and Power BI.
- Worked with Oracle, SQL, Twilio, and Zendesk connectors to integrate data with Power BI.

Intern | Algonation, Mumbai, India **Dec 2014 – Feb 2015**

- Gathered client requirements and created a MySQL data model for a healthcare project.
- Created a website using HTML5, CSS3, PHP, JavaScript, and the CodeIgniter framework.

EDUCATION

University of Texas at Arlington, USA — Master of Science in Computer Science, 2019
K. J. Somaiya College of Engineering, India — Bachelor of Information Technology, 2015

ACADEMIC PROJECTS (MASTER'S, UT ARLINGTON)

Movie Suggestion System

- Created a Flask application in Python 3.6 using NumPy, NLTK, and Pandas for word processing, including stemming and lemmatization.
- Processed a database of 1.5M rows to provide genre-based prediction results and deployed the application on Google Cloud Platform.

Online Advising Simulation

- Developed a client-server multi-threaded application for an online student-teacher advising system.
- Implemented message queuing using the producer-consumer design pattern.